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Section 6.5

37. $m_{Y_1}(t) = E[e^{tY_1}] = \sum_j e^{tj} p(Y_1=j) = \sum_{j=0}^1 e^{tj} (Y_1=j)$
 $= e^{t \cdot 0} \cdot q + e^{t \cdot 1} \cdot p = pe^t + q$ ✓

4. Let $W = Y_1 + Y_2 + \dots + Y_n$ in Bernoulli.

$$m_W(t) = m_{Y_1}(t) \cdot m_{Y_2}(t) \cdot \dots \cdot m_{Y_n}(t)$$

$$= (pe^t + q)^n$$

c. $W \sim \text{Bin}(n, p)$

4b Let $Y \sim \text{Gamma}(\alpha = n/2, \beta)$, $W = 2Y/\beta$

$$m_Y(t) = (1 - \beta t)^{-\alpha}$$

$$m_W(t) = E(e^{tW}) = E[e^{t(2Y/\beta)}] = E[e^{\frac{2}{\beta}tY}] = m_Y(\frac{2}{\beta}t)$$

$$\Rightarrow (1 - \beta(\frac{2}{\beta}t))^{-\alpha} = (1 - 2t)^{-n/2}$$

$$Y \sim \text{Chi-square}(n)$$

50 Let $Y \sim \text{Bin}(n, p)$ [show $n - Y \sim \text{Bin}(n, q)$]

$$m_Y(t) = (pe^t + q)^n$$

$$m_{n-Y}(t) = E[e^{t(n-Y)}] = E[e^{nt - tY}] = e^{nt} E[e^{-tY}]$$

$$= e^{nt} \cdot m_Y(-t)$$

$$= e^{nt} (pe^{-t} + q)^n$$

$$= (e^t)^n \cdot (pe^{-t} + q)^n = (pe^{-t} + qe^t)^n = (p + qe^t)^n$$

$$n - Y \sim \text{Bin}(n, q)$$

54a $m_{Y_i}(t) = e^{\lambda_i(e^t - 1)} = \exp[\lambda_i e^t - \lambda_i]$

$$m_{\sum_{i=1}^n Y_i}(t) = m_{Y_1}(t) \cdot m_{Y_2}(t) \cdots m_{Y_n}(t)$$

$$= e^{\lambda_1(e^t - 1)} \cdot e^{\lambda_2(e^t - 1)} \cdots e^{\lambda_n(e^t - 1)}$$

$$= \exp[\lambda_1 e^t - \lambda_1] \cdot \exp[\lambda_2 e^t - \lambda_2] \cdots \exp[\lambda_n e^t - \lambda_n]$$

$$= \exp[\sum_{i=1}^n \lambda_i e^t - \sum_{i=1}^n \lambda_i]$$

$$P_{\sum_{i=1}^n Y_i}(y) = \frac{(\sum_{i=1}^n \lambda_i)^y}{y!} e^{-\sum_{i=1}^n \lambda_i}, \quad y = 0, 1, \dots$$

59. Let $Y_i \sim \text{Chi-square}(\nu_i)$, $U = Y_1 + Y_2$

$$m_{Y_1}(t) = (1 - 2t)^{-\nu_1/2}$$

$$m_U(t) = m_{Y_1}(t) \cdot m_{Y_2}(t)$$

$$= (1 - 2t)^{-\nu_1/2} (1 - 2t)^{-\nu_2/2}$$

$$= (1-2t)^{-\nu_1/2 - \nu_2/2}$$

$$= (1-2t)^{-\frac{\nu_1 + \nu_2}{2}}$$

$$U \sim \text{Chi-square}(\nu_1 + \nu_2)$$



Section 6.7